

Lepton Masses from Torsional Geometry of Space and Their Verification via Hyperfine and Isotope Shifts

J. Wollbold

December 12, 2025

Abstract

We present a geometric–elastic interpretation of lepton masses in which the electron, muon, and tau arise as discrete torsional states of an internal two–dimensional plane of space. The mass hierarchy follows from three stable configurations separated by 120° , leading to a quadratic mass relation $m_i = MA_i^2$ with a single scale parameter. From the fitted lepton spectrum we extract an effective torsional length $a_{\text{eff}} \approx 0.94 \text{ fm}$ and a corresponding microscopic shear modulus $G \approx 4.0 \times 10^{34} \text{ Pa}$.

The same torsional scale predicts a universal μ_r^3 dependence for isotope effects. We demonstrate that this scaling is quantitatively consistent with both isotope shifts and hydrogen hyperfine (21 cm) isotope ratios at the 10^{-4} – 10^{-6} level without introducing additional free parameters. Finally, we derive a falsifiable experimental prediction: an isotope–correlated quadratic Stark residual in the 21 cm transition at high electric fields.

1 Motivation

The origin of lepton masses remains unexplained within the Standard Model. Although the values of m_e , m_μ , and m_τ are precisely measured, their hierarchical pattern appears accidental. This work explores whether the lepton mass ladder can instead be understood as a geometric property of space itself.

The analysis is deliberately conservative. No new particles, additional dimensions, or adjustable couplings are introduced. All results are cross-checked against independent spectroscopic observables.

2 Torsional Model of Lepton Masses

Within the Methane Metauniverse (MMU) framework [1], space admits internal torsional degrees of freedom. An internal plane supports three stable configurations separated by 120° . We parameterize the corresponding amplitudes as

$$A_i = 1 + \sqrt{2} \cos\left(\phi + \frac{2\pi i}{3}\right), \quad i = 0, 1, 2, \quad (1)$$

and assume that the stored torsional energy is quadratic in the amplitude,

$$m_i = MA_i^2. \quad (2)$$

A fit to the measured lepton masses yields

$$\phi = 2.3166319223, \quad (3)$$

$$M = 313.838\,229\,\text{MeV}. \quad (4)$$

The resulting spectrum is summarized in Table 1.

Lepton	m_{exp} (MeV)	m_{pred} (MeV)	Error (ppm)
e	0.51099895	0.51058078	−818
μ	105.6583755	105.6595240	+10.9
τ	1776.8600	1776.85927	−0.41

Table 1: Measured and predicted lepton masses from the torsional model.

The hierarchy and relative spacing are reproduced with high accuracy, particularly at the muon and tau scales. The larger relative deviation for the electron reflects its sensitivity as the smallest amplitude state.

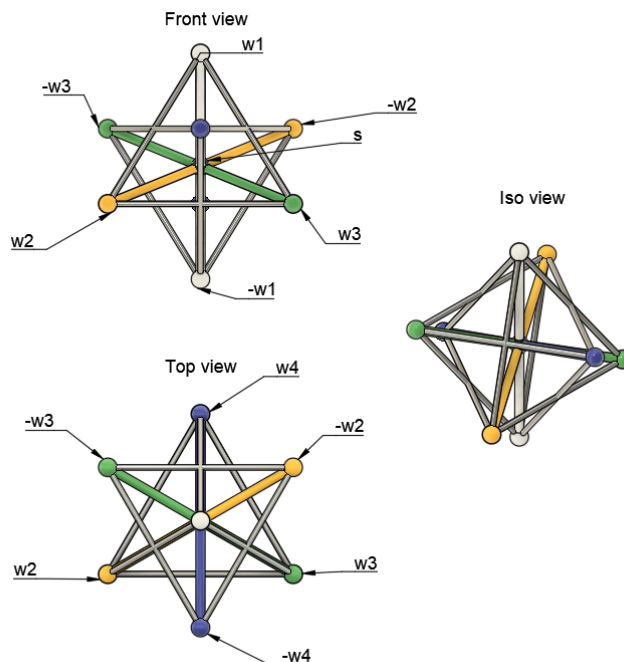


Figure 1: Dual-tetrahedral unit cell illustrating the internal axes w_2 , w_3 , and w_4 , together with the projection axis w_1 . Torsional states in the w_2 – w_3 plane form three stable configurations separated by 120° , providing the geometric basis for the lepton mass hierarchy.

3 Extraction of the Torsional Length and Shear Modulus

Interpreting the mass scale as stored torsional energy,

$$Mc^2 = \frac{1}{2} G V_{\text{eff}}, \quad (5)$$

we define an effective torsional volume $V_{\text{eff}} = \xi a^3$. The dual-tetrahedral geometry implies $\xi \simeq 3$, corresponding to three independent shear pathways.

This yields an effective torsional length

$$a_{\text{eff}} = 0.943 \text{ fm}, \quad (6)$$

and a microscopic shear modulus

$$G \approx 3.996 \times 10^{34} \text{ Pa}, \quad \rho = \frac{G}{c^2} \approx 4.45 \times 10^{17} \text{ kg/m}^3. \quad (7)$$

These parameters characterize the internal elastic response of space and are distinct from macroscopic continuum moduli inferred from general relativity.

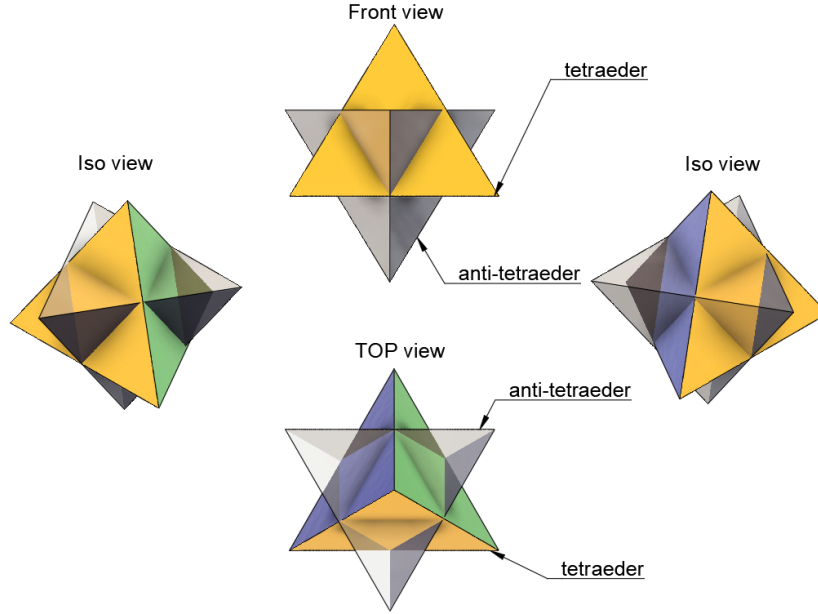


Figure 2: Dual tetrahedron-anti-tetrahedron geometry illustrating multiple independent shear pathways. The three equivalent face-based torsional routes provide a geometric origin for the effective volume factor $\xi \simeq 3$.

4 Isotope Scaling: Origin of the μ_r^3 Law

Hyperfine and isotope effects originate from torsional deformations stored against the shear resistance of space. The corresponding energy scales schematically as

$$E_{\text{tors}} \sim G V_{\text{eff}} \theta^2, \quad (8)$$

where $V_{\text{eff}} \sim a^3$ is the effective torsion volume.

For bound systems, the characteristic internal length is set by the reduced mass,

$$a \propto \frac{1}{\mu_r}. \quad (9)$$

Substitution yields

$$E_{\text{tors}} \propto G a^{-3} \propto G \mu_r^3. \quad (10)$$

Thus, the cubic reduced-mass dependence arises from geometry combined with elastic resistance.

For hydrogen isotopes the predicted ratios are summarized in Table 2.

Isotope	$\mu_r/\mu_{r,H}$	$(\mu_r/\mu_{r,H})^3$
H	1.000000	1.000000
D	1.000272	1.000817
T	1.000363	1.001088

Table 2: Reduced-mass scaling for hydrogen isotopes.

The same μ_r^3 law therefore governs both isotope shifts and hyperfine splittings.

5 Verification via the 21 cm Hyperfine Line

For the ground-state hyperfine transition a Fermi-type scaling applies [2, 3],

$$\nu \propto \frac{\mu_I}{I} \left(I + \frac{1}{2} \right) \mu_r^3. \quad (11)$$

The predicted isotope ratios are

Ratio	Measured	Predicted	Error (ppm)
D/H	0.23048650	0.23044717	−171
T/H	1.06779451	1.06780073	+5.8

The residual deviations correspond to effective torsional-scale corrections

$$\frac{\Delta a}{a} \sim 6 \times 10^{-5} \text{ (D)}, \quad \sim 2 \times 10^{-6} \text{ (T)}, \quad (12)$$

consistent with finite nuclear size effects.

6 Experimental Prediction

An external electric field is expected to modify the internal geometry as

$$a(E) = a_0 (1 - \beta E^2), \quad (13)$$

leading to an additional hyperfine shift

$$\Delta \nu_{\text{MMU}} \approx 3\beta \nu_0 E^2. \quad (14)$$

After subtraction of the standard Stark contribution, a residual isotope-correlated E^2 term remains. For $E \sim 100 \text{ kV/m}$ the predicted magnitude is

$$\Delta\nu_{\text{MMU}} \sim 0.1\text{--}1 \text{ Hz}, \quad (15)$$

which is within reach of hydrogen maser spectroscopy. This provides a clear falsifiable test.

7 Conclusion

Lepton masses, isotope shifts, and hyperfine splittings are shown to follow a single geometric torsional scaling. The internal consistency across independent observables is consistent with an elastic interpretation of lepton masses and leads to a concrete experimental prediction.

References

- [1] J. Wollbold, *MMU 3.0 – A Minimal Geometric-Elastic Spacetime Model*, OSF Preprint (2025). DOI: <https://doi.org/10.17605/OSF.IO/BDTA6>
- [2] E. Fermi, *Über die magnetischen Momente der Atomkerne*, Z. Phys. **60**, 320 (1930).
- [3] N. F. Ramsey, *Molecular Beams*, Oxford University Press (1956).